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Mapping party systems over time: a spatial distribution of ideological competition in Chile

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ABSTRACT

This article introduces a novel method to map party systems over time, addressing the challenge of assigning comprehensive spatial ideological scores to coalitions, parties, and independents. Combining public opinion polls, manifesto analysis, and expert surveys, this method evaluates fixed and variable standard deviation models to score distinct 4,157 electoral lists from 5,195 district-level elections across 27 elections in 12 Chilean electoral systems. It proposes a synthetic data approach to simulate party positions within political family constraints, producing a high-density region map and a political lineage chart that visualize ideological evolution over time. Robustness checks, including diagnostics with controlled random noise and variance adjustments, confirm the reliability of these results. Key findings reveal an earlier-than-expected trend of fragmentation, a leftward drift among left-leaning parties, and an asymmetrical party system where fragmentation affects the left while polarization intensifies on the right. This study underscores the importance of expanding within-system estimators to enhance reliability and concludes with research questions that the resulting dataset can address. These include constituency-level shifts in ideological positioning, fragmentation linked to specific institutional reforms, and the dynamics of localized polarization within multi-party systems.

1. Introduction

Estimating the ideological scores of political parties has been a central concern in political science since the rise of modern democracy.¹ Conceiving and categorizing parties according to their ideology has contributed to the understanding not only of the public policy process but also to the development of party system literature.² Traditional research suggests that party position distributions can be estimated using political manifestos, roll call votes, public opinion polls, or expert surveys.

However, while all of these methods are useful to understand where parties or candidates position themselves, none allow for the mapping of a full political system. This article presents proposes a method to score all coalitions, parties and independents competing in elections using a partial non-parametric synthesizing method and shows some preliminary results. It argues in favor of full-system scoring and highlights some of

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the areas that can benefit from such a process. It acknowledges limitations but suggests a manner in which the reliability of the estimators can be maximized.

The remainder of this article is structured as follows. The next section reviews foundational literature on spatial models and other methods for estimating party positions, highlighting their limitations in capturing the complexity and dynamics of ideological competition. The third section defines the specific research objective, explains the choice of Chile as a case study, and describes the district-level data used for ideological scoring. The fourth section details the methodological approach, covering the creation of the original and synthetic datasets and describing the development of both fixed and variable standard deviation models to assign scores across coalitions, parties, and independents. The fifth section presents the results, including high-density region maps and a political lineage chart that illustrate the ideological evolution within Chile's party system. Finally, the sixth section concludes with suggested research questions on the dataset's potential applications, including exploring the impact of institutional reforms on ideological shifts, fragmentation, and polarization.

2. Previous research

The theoretical foundation for mapping ideological positions in party systems is rooted in Downs' spatial model of party competition.³ According to Downs, parties strategically position themselves along an ideological spectrum to maximize electoral support, with voters choosing parties based on perceived ideological proximity. The model envisions both parties and voters on a continuum, typically from left to right, representing ideological stances. The simplicity and rational-utility assumptions of Downs' model have cemented it as a cornerstone in studies of electoral competition and party systems.

From this, researchers have explored the complexities of spatial competition, adding dimensions that reflect real-world political dynamics. Shepsle generalizes Duncan Black's median voter theorem, underscoring the role of institutional structures in shaping equilibrium outcomes in electoral competition.⁴

Hinich and Pollard propose a multidimensional approach to spatial theory, recognizing that parties and voters consider multiple issues that cannot always be reduced to a single left-right axis.⁵ This added complexity acknowledges that ideological positioning often involves balancing commitments across a range of policy dimensions.

Stokes expands on this by challenging the notion of fixed ideological stances, proposing instead that party competition frequently revolves around valence issues – outcomes that all voters desire (such as economic stability or public safety) but for which parties compete based on perceived competence.⁶ This shift emphasized the need for ideological mapping to include both policy and valence dimensions to accurately capture voter-party alignment.

In turn, Budge introduces a temporal aspect to the spatial theory, highlighting that party competition is fluid and adapts over time to shifting societal concerns.⁷ In this account, party positioning and voter preferences are not static; they evolve as parties respond strategically to new issues or recalibrate in response to competitors – which underscores that ideological mapping must be adaptable, capturing not only the multi-dimensional space of issues but also the changing landscape of party competition over time.

These foundational contributions underscore the limitations of traditional one-dimensional models, pointing toward a need for mapping methods that account for the multidimensionality, temporal dynamics, and non-policy factors inherent in real-world party competition. Together, they suggest that empirical approaches to ideological mapping must be flexible enough to accommodate these theoretical complexities.

In practice, several empirical methods have been developed to meet this challenge and map ideological positions.⁸ One widely used approach is manifesto content analysis, where political programs are systematically scored based on a common coding scheme.⁹ This method associates programmatic positions with specific parties, allowing analysts to estimate relative ideological distances.¹⁰ However, issues such as coder bias, selection limitations, and interpretive differences can affect the reliability of text-based mappings, especially when political messages are complex or nuanced.¹¹

Another prominent approach involves legislative roll-call scaling methods, such as DW-NOMINATE, which position parties based on the voting behavior of their legislators.¹² While effective in contexts where parties are represented in the legislature, this method inherently excludes parties without legislative seats, potentially overlooking important ideological actors within the broader party system.¹³ Additionally, scaling methods may be limited by the inherent constraints of the legislative context, where voting often aligns with party discipline rather than true ideological positioning. There are question as to whether Bayesian methods may ultimately be more accurate.¹⁴

Public opinion surveys also offer a means to map ideological spaces by asking respondents to place parties on an ideological continuum.¹⁵ This approach, as seen in resources like the World Values Survey and the Latin American Public Opinion Project, provides a broad view of public perceptions of party positions. However, its reliability can be compromised when respondents lack familiarity with ideological scales or specific parties, potentially skewing average placements and leaving gaps in ideological mappings for lesser-known parties.¹⁶

Expert surveys represent another approach, where specialists score political parties on ideological scales.¹⁷ Resources like the Chapel Hill Expert Survey and Varieties of Democracy (V-Dem) use this method to provide highly reliable scores, leveraging experts' in-depth knowledge.¹⁸ Yet, this approach also has limitations, including potential subjective biases and a lack of information for less prominent parties. Furthermore, it remains uncertain whether aggregating individual party scores effectively captures the overall ideological balance of the party system, as any omission introduces potential biases in system-wide mappings, as it does not solve the issue of human-related subjective nuances or lack of information for lower (less prominent) levels of electoral competition.¹⁹

Together, these methods underscore both the demand for and the challenges of comprehensive ideological mapping. To capture the full landscape of party competition, methods must account for ideological diversity across parties, potential biases in data sources, and the inclusion of both major and minor political actors. This study addresses these challenges by introducing a synthetic data approach that builds on the strengths of traditional mapping methods while addressing their limitations. By integrating multi-dimensional flexibility and temporal adaptability, this approach provides a robust framework for mapping ideological positioning in complex party systems, aligning with

theoretical insights from foundational spatial models and advancing the empirical toolkit for studying party competition.

3. Objective, research question and case selection

Building on foundational theories of spatial party competition outlined above, this study seeks to address a core research question: *How can we comprehensively map ideological positioning within a complex, multiparty system over time?* Specifically, we focus on Chile's political landscape, a setting characterized by fluid coalitions, diverse electoral systems, and fluctuating ideological alignments, which pose unique challenges to traditional one-dimensional or static mapping approaches. The goal is to develop a method that not only captures ideological positions at individual points in time but also tracks shifts in positioning across electoral cycles, thus reflecting the dynamic and multidimensional nature of party competition.

This research question emerges from limitations in previous mapping approaches, which often rely on static or single-dimensional methods that struggle to capture the evolving ideological spectrum in multiparty systems. As discussed, foundational models like Downs' spatial theory,²⁰ Shepsle's strategy of ambiguity,²¹ and the multidimensional expansions by Hinich and Pollard highlight the complexity of ideological positioning.²² These theories suggest that party competition involves a blend of ideological positioning, tactical uncertainty, and adjustments based on issue valence and temporal shifts. In line with these insights, our research question underscores the need for a method that is adaptable to these complexities, especially in a context like Chile's, where coalitions can shift parties and ideological cohesion can vary significantly across elections and over time.

To answer our research question, we introduce a novel dataset and a two-stage methodological approach. The first stage involves constructing a comprehensive list of all electoral lists observed across Chile's diverse election types and levels. This initial compilation is essential for establishing a complete baseline of coalition, party and independent positions across multiple electoral cycles. The second stage involves generating synthetic data that allows for simulating ideological shifts and variations within these electoral lists over time. This synthetic dataset, built upon the observed data, incorporates controlled random noise to reflect natural ideological variation within coalitions while maintaining core ideological groupings. By simulating plausible ideological positions for each electoral list across time, we create a longitudinal view of party competition that aligns with theoretical models emphasizing multidimensionality, strategic ambiguity, and the impact of temporal changes on party positioning.

Chile is chosen as a case study due to its complex and multiparty structure, which provides a rich context for testing this method. The dataset encompasses electoral returns from 5,195 district-level competitions, covering 27 elections across seven distinct electoral systems, and offers a broad empirical foundation to explore ideological positioning within a multiparty system. This includes data from Municipal, Local Councilor, Mayoral, Deputy, Constitutional Convention, Regional Councilor, and Senatorial elections, allowing for analysis across multiple governance levels. These data, compiled into a master dataset of 4,157 unique observations, provide a granular view of ideological

alignments within coalition-party, coalition-independent, party-party, party-independent, and independent-independent combinations.

In scoring these electoral lists, we draw on a diverse set of sources, including country-specific political science literature, press reports, public opinion polls, elite surveys, and political manifestos. This approach ensures that the resulting ideological positions are grounded in both historical context and current political realities, increasing the validity of the estimator. This method also allows for adaptability, so that future studies can refine or expand upon these scores as new information or analytical tools become available. Our goal is to ensure that this estimator not only provides continuity with existing scholarship but also serves as a reliable foundation for future research on ideological shifts in Chile's party system.

This study is the first to apply such a comprehensive mapping approach to Chile's political landscape. Past studies on ideological positioning in Chile often rely on individual methods, such as manifesto analysis, roll-call scaling, or public opinion surveys, which tend to focus on particular parties, coalitions, or electoral moments.²³ These approaches, while valuable, lack the breadth and continuity needed to capture the ideological evolution of the entire party system over time. By combining a detailed empirical dataset with a synthetic approach, this study bridges these gaps, offering a consistent and longitudinal mapping that accounts for both established parties and new political actors within the broader system.

The resulting estimator provides a score for all electoral lists, encompassing coalition and independent entities, with the flexibility to account for shifts in relative positioning over time. By using a synthetic dataset grounded in empirical data, we can model ideological distances within expected margins of error, thus enabling a more nuanced understanding of ideological alignment and competition in Chile's multiparty democracy. The estimator's simplicity and adaptability make it suitable for ongoing updates, allowing researchers to recalibrate and expand the data as new elections and political changes occur.

4. Methods

We now turn to the two-stage synthetic approach. This sequential-tier method is designed to capture ideological positions over time across the entire party system. The first stage involves scoring the original data, which provides ideological estimates for all coalitions, parties, and independents. In the second stage, a synthetic dataset is generated, building upon the scored data to produce a more robust, probabilistic mapping of ideological shifts. Together, these stages offer a comprehensive, adaptable, and replicable solution for analyzing ideological dynamics.

Traditional methods such as manifesto analysis, legislative roll-call scaling, public opinion surveys, and expert evaluations have made significant contributions to our understanding of ideological positions but are limited in their applicability for longitudinal studies. Manifesto analysis, for instance, is instrumental for capturing programmatic positions but is constrained by issues such as selection bias, human error, and inconsistent classifications, especially when ideological content is sparse or fragmented over time. Legislative roll-call scaling methods, like DW-Nominate, apply only to parties with legislative representation, thereby excluding influential

yet non-represented parties. This is a critical limitation in a multiparty system like Chile's, where ideological diversity often extends beyond the legislature. Public opinion polling provides valuable insights but is prone to inaccuracies when respondents lack familiarity with ideological scales or party structures. Finally, expert surveys offer reliability but often fail to score minor parties or independent candidates in local-level competitions.

Our synthetic approach mitigates these limitations by first establishing a baseline with scored original data, which integrates multiple sources – opinion polls, expert assessments, and manifestos – to estimate ideological positions. This initial scoring step ensures that each coalition, party, and independent candidate has an ideological position assigned, regardless of their level of legislative representation. By covering all political entities, this stage allows for a comprehensive mapping that circumvents the structural exclusions of traditional methods.

The second stage involves generating synthetic data based on the initially scored dataset. This synthetic dataset introduces random noise calibrated to observed variances within major political families, which enhances robustness across the ideological spectrum. Repeated simulations and averaging create a longitudinal, probabilistic dataset, allowing us to capture subtle ideological shifts over time with increased reliability. This two-stage process also addresses potential human biases and inconsistencies by averaging across multiple simulations, thus producing a more stable and consistent dataset.

Compared to existing approaches, this two-stage model brings several distinctive advantages. First, it provides a more nuanced tracking of ideological shifts, especially valuable in Chile's political context, where ideological realignments can be gradual and complex. Second, the synthetic model is adaptable, allowing for updates as new data becomes available, making it suitable for analyzing changes over time, including those resulting from electoral reforms. Finally, this approach enables spatially comprehensive visualizations of ideological competition, illustrated through high-density region maps that show ideological distributions and their evolution over time, surpassing the capabilities of static or fragmented analyses.

4.1. Original scoring process

The following steps were used to score coalitions, parties, and independents on an ideological scale ranging from 0 (far left) to 10 (far right):

- (1) A list of all coalitions, political parties, and independents that competed in an election was generated from the master dataset, yielding a total of 4,157 observations, each representing a unique election-coalition-party combination.
- (2) Traditional coalitions (those participating in elections since 1989) were scored using averages derived from public opinion surveys, specifically the CEP and CERC polls.²⁴ These scores were cross-validated against existing studies to ensure consistency.²⁵
- (3) Both fixed (FSDM) and variable (VSDM) standard deviation models were fit to score parties and independents within coalitions based on prior studies indicating relatively high intercoalition discipline that may vary depending on the size of the coalition.²⁶

- (4) Traditional parties (those operating within established coalitions) were assigned scores within this defined deviation range.
- (5) Other coalitions (those formed post-1989 or participating in only one or a few elections) were scored using available public opinion surveys or, when necessary, secondary sources like press reports, which provided indirect estimates of ideological distance relative to traditional parties. Reviewed press sources included but were not limited to national newspapers such as *La Tercera* and *El Mercurio*.²⁷
- (6) Other parties (those competing within less established or newer coalitions) were scored within the same standard deviation range.
- (7) Independent candidates within coalitions were assigned the same score as their supporting coalition.
- (8) Independent candidates within parties were given the same score as their endorsing party.
- (9) Independents running without any coalition or party endorsement were assigned a neutral score of '5'.
- (10) Finally, scores were verified to ensure consistency with previous findings across sources.

Table 1 shows the resulting distribution of scores in a generic categorization from far left (0) to far right (10) after applying the method described in the previous section. Table A1 (in the Appendix) provides a more detailed breakdown, displaying ideological scores by type of election, electoral system, and political family in Chile from 1989 to 2023. It shows that the center category (5), reserved for independents, accounts for the highest number of cases within any range. Because this bin spans from 4.5 to 5.5, other coalitions and parties are also included, though their relative weight is minor. When disaggregated, there are 3,378 independents and 48 coalition-party, coalition-independent, or party-party combinations (total = 3,426).

Table A2 in the Appendice summarizes the distribution of types of lists competing in elections in Chile. It shows three distinct classifications: full independents (candidates with no party endorsement), independents endorsed by a party (candidates running with indirect support from a political party), and party lists (formal party candidates). This categorization allows for a nuanced understanding of how independents and party-affiliated candidates are distributed ideologically. This breakdown is particularly relevant

Table 1. Distribution of electoral lists and total votes by political family in Chile, 1989–2023.

Political family	N	Total votes
Far Left [0.00–1.00]	47	7,311,139
Left [1.01–3.49]	315	22,626,685
Center-Left [3.50–4.5]	201	57,706,325
Center (includes Independents=5) [4.51–6.49]	3,426	22,208,073
Center-Right [6.50–7.50]	86	31,708,205
Right [7.51–8.99]	69	21,459,962
Far Right [9.00–10.00]	13	10,596,640
Total	4,157	173,617,029

Source: Author. Note. Bins are referential (see CEP public opinion survey), roughly following a normal distribution.

in proportional representation elections, such as legislative contests, where independents must compete against party lists on an equal footing. Although there is a high number of independent candidates, their cumulative weight in votes remains limited. Excluding the 48 coalition and party-associated lists, independents only account for slightly more than 7 million out of nearly 174 million total votes tallied (approximately 4%).

4.2. Synthetic data generation

The resulting dataset from the fixed standard deviation model (FSDM) scoring process is transformed to simulate potential biases and errors, producing a synthetic dataset that provides a more robust estimation of ideological positions.²⁸ This synthetic data generation approach incorporates several steps designed to replicate the variability observed within ideological families while preserving the underlying structure of the original data.

- (1) Initial Data Preparation and Random Noise: We begin by importing the original dataset and calculating the percentage of votes each electoral list received in each election. To introduce controlled variability, random noise is added to the original scores, allowing for slight deviations that reflect natural ideological variation within each category.²⁹
- (2) Setting Parameters for Simulations: We define the number of simulations to be conducted (in this case, 1,000) to ensure that the synthetic dataset captures a broad range of plausible ideological positions. This high number of simulations allows for a probabilistic mapping that minimizes the influence of outliers and provides a stable average score.³⁰
- (3) Grouping by Political Families: Political entities are grouped into ideological families (e.g. Far Left, Left, Center Left, Center, Center Right, Right, Far Right), considering the nature of coalition competition and the effective number of parties in Chile.³¹ For each family, we calculate the standard deviation, minimum, and maximum values of the ideological scores. These parameters help simulate ideological positions within realistic bounds, reflecting typical variance observed within each political family.
- (4) Generating Synthetic Scores: Using a custom function, we generate synthetic ideological scores by introducing random variation around the family-specific mean, constrained by the calculated standard deviation. This approach respects the ideological 'boundaries' of each political family while allowing for realistic variability. The synthetic scores are generated multiple times across simulations, producing a distribution that reflects the ideological positioning for each party over time.
- (5) Averaging Synthetic Scores: After all simulations are completed, the synthetic scores are averaged to create a final synthetic ideological position for each observation. This averaging process mitigates the influence of extreme values and provides a more stable and representative score for each party or coalition.
- (6) Diagnostic and Validation Checks: To validate the reliability of the synthetic data, we perform a series of diagnostic checks. These include comparing synthetic scores to original scores to assess consistency, checking for normality in score distributions, and testing for outliers and heteroscedasticity. Additionally, we

generate visual comparisons (e.g. kernel density plots, regression diagnostics) to confirm that the synthetic data maintains the core patterns observed in the original data while introducing controlled variability.

Figure A1 and Figure A2 in the Appendice show the original dataset scores against the synthetic data scores, and the synthetic data scores against the difference of both scores.³² They show that the magnitude of the deviations is small (lower than 1.0 point, on the eleven-point scale, in either direction), and normally distributed around the mean, indicating the absence of heteroscedasticity.

Figures A3-A8, also in the Appendice, show diagnostics after regressing synthetic data scores on the original data scores (including residual vs. fitted distributions, normal Q-Q analysis, scale location placements, Cook's distance, residuals vs leverage, and Cook's distance vs. leverage).³³ Generally, they show no indication of major outliers, and provide evidence in favor of consistency of the synthetic data – which was expected from the designated within group error matrix (standard deviations tied to political families).

Finally, Figure A9 and Figure A10 show kernel density distributions of ideological scores for both datasets (synthetic and original) side by side, showing that there is no apparent difference across the continuous horizontal axis.³⁴

5. Findings: exploring party systems over time

Figure 1 illustrates the ideological clustering within Chile's party system using a maximum likelihood estimation for high-density regions.³⁵ Full independents are removed to isolate party competition patterns. The visualization identifies areas where ideological clusters are most concentrated over time, allowing us to observe shifts in party alignment and entry. The map shows a notable uptick in the number of parties participating in the political system beginning around 2009, indicating signs of fragmentation. This pattern of increased party entries and clustering suggests shifts in competition dynamics, with new parties or coalitions joining the ideological spectrum.

Interestingly, the observed fragmentation begins earlier than is generally acknowledged in the literature.³⁶ While the prevailing consensus attributes significant party system expansion to the 2015 electoral reform – which formally reduced barriers for new parties and independents,³⁷ the data here reveals an emerging fragmentation trend as early as 2009. This early diversification within the party system contrasts with typical accounts that primarily link fragmentation to high-profile legislative reforms, particularly those influencing the structure of legislative elections. Our findings suggest that fragmentation may have been underway before these reforms, hinting at underlying sociopolitical shifts that traditional analyses may have overlooked.

The early signs of fragmentation are likely visible because of the use of district-level data and the inclusion of lower-profile elections, which include smaller parties with low vote shares. By examining these often-overlooked elections, we were able to capture shifts that would not appear in analyses focused solely on high-profile legislative elections. This broader dataset allows for a more comprehensive view of ideological changes across the full political spectrum, showing fragmentation patterns that emerge earlier than previously thought.

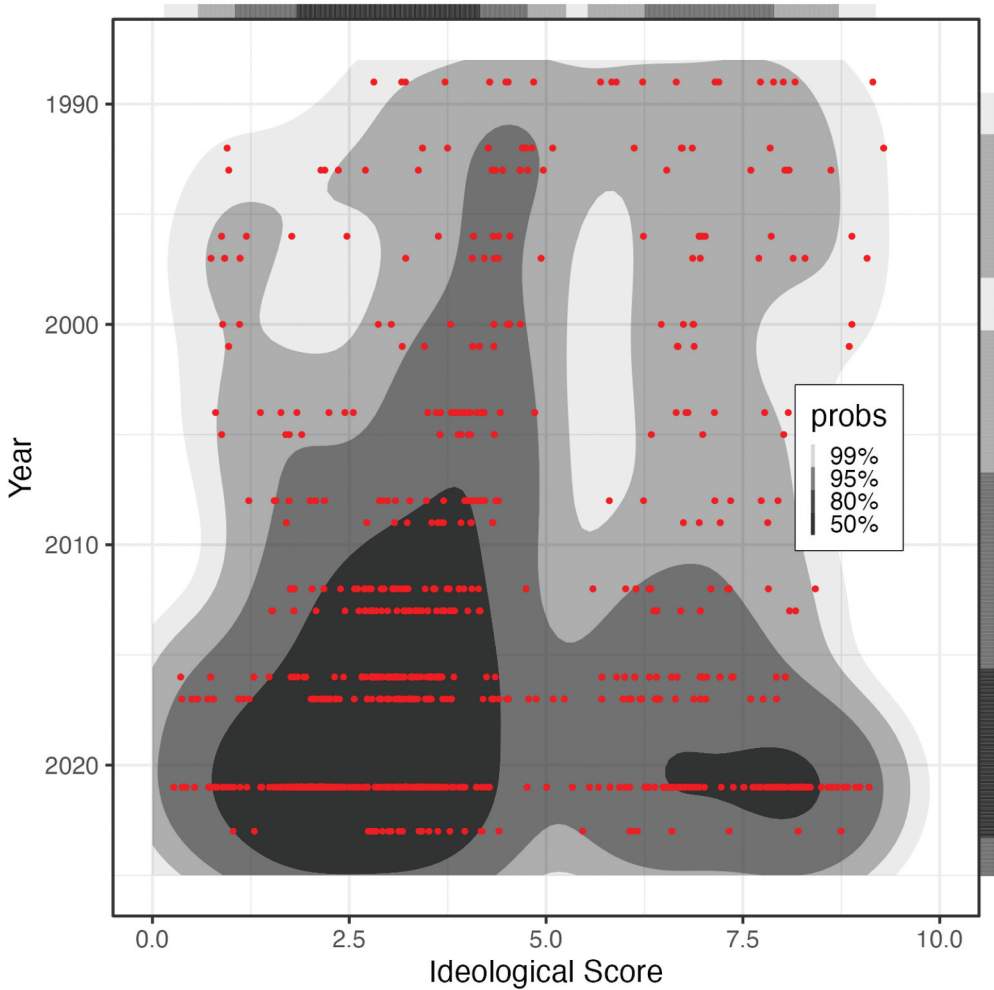


Figure 1. Patterns of ideological fragmentation in Chile's party system over time. Source: Author. Note: Data based on spatial distribution high density regions.

Building on the observation that a high-density region began forming to the left of center since about 1993, it is useful to further investigate general drifting patterns over time. Figure 2, alongside regression results detailed in Table A3 in the Appendix, provides insights into this question, suggesting that as more parties entered the system, there was an observable trend toward a leftward shift in ideological positioning. This reflects potential dynamics in spatial competition, where parties may have been responding to voter distributions and strategic positioning pressures related to the long-term incumbency effects of the center left coalition in power.³⁸

The regression analysis, focusing on left-leaning parties (those with ideological scores below 5), reveals a statistically significant negative trend in positioning over time. Specifically, the left-wing regression model shows a highly significant negative coefficient ($b = -0.04$, $p < 0.001$), indicating that left-wing ideological scores tend to decrease with each electoral cycle. This suggests a pattern of increasing fragmentation

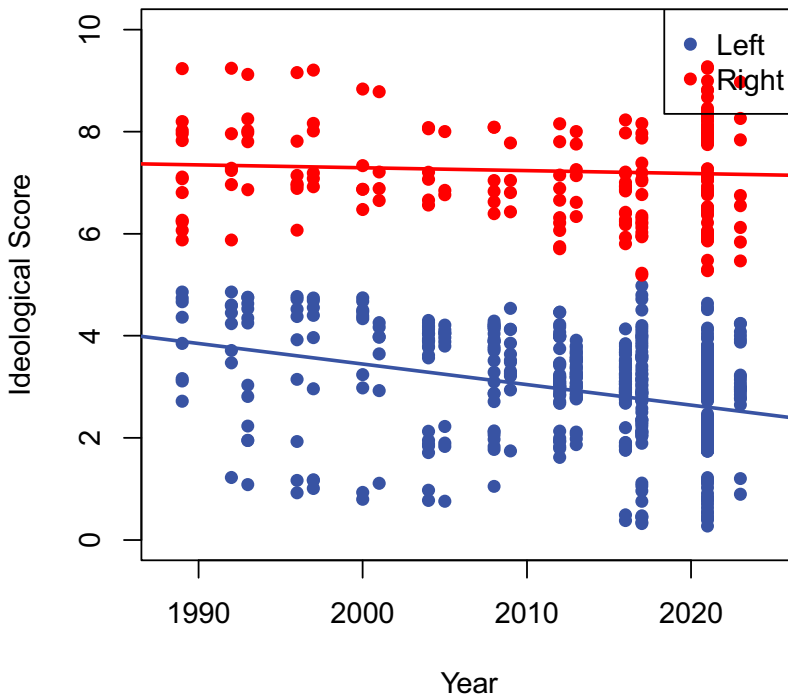


Figure 2. Trends in ideological shifts among left and right leaning parties over time. Source: Author. Note: Data based on regression analysis shown in Table A3. Points above 5 represent right of center; points below 5 represent left of center.

or a progressive shift toward more extreme left positions as new parties join the system or as existing left-leaning parties position themselves further from the center. This leftward drift aligns with the observed increase in ideological competition within the left, reflecting a trend of ideological diversification and intensification of competition among left-leaning parties.

In contrast, the regression analysis for right-wing parties, defined as those with ideological scores above 5, shows no significant trend over time. The right-wing model presents an almost flat slope with a high p-value, indicating no statistically significant change in ideological positioning among right-leaning parties. This suggests a level of stability and cohesion within the right, with limited ideological drift or fragmentation. The stability observed here could point to a tendency among right-leaning parties toward consolidation, where larger, established parties maintain dominance, reducing the likelihood of fragmentation into smaller, ideologically similar groups.

These findings reveal a discrepancy in the ideological evolution of left- and right-leaning parties within Chile's political system. While left-leaning parties exhibit a clear pattern of increasing ideological diversity and competition, right-leaning parties remain relatively stable and consolidated. This difference in ideological development highlights the varying strategies and structural dynamics within each political family, with potential implications for voter alignment and coalition-building in Chile's multiparty democracy.

To further explore the observed ideological dynamics, Figure 3 presents a spatial competition map over time, capturing both parties and independents.³⁹ In this Figure,

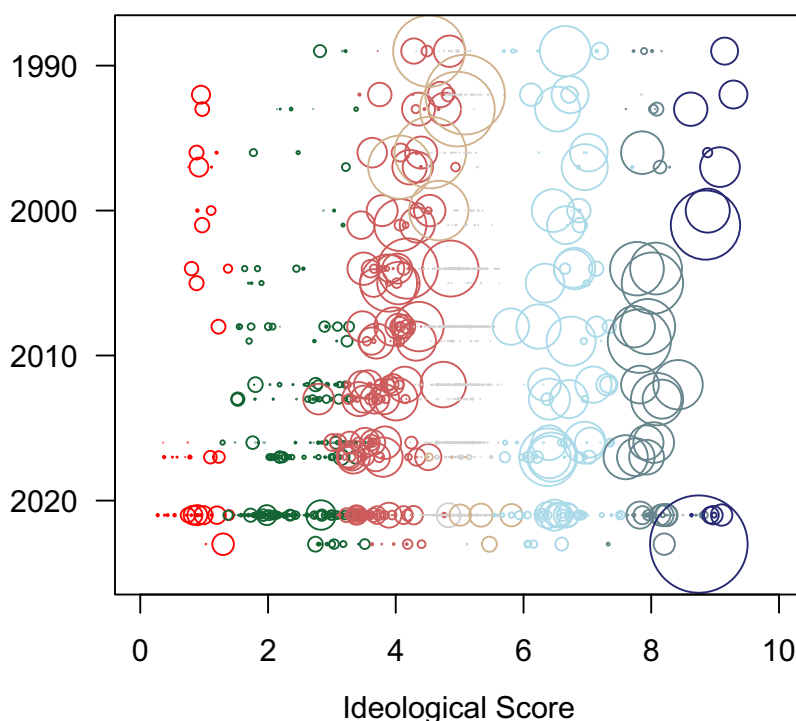


Figure 3. Party system asymmetry across political spectrum over time. Source: Author. Note: Data weighed by vote percentage. Colors based on categories referenced in Table 1.

each circle's size reflects the percentage of votes obtained by each party per election, with larger circles indicating a more substantial share of votes.⁴⁰

The spatial distribution highlights an asymmetry in how fragmentation and polarization are developing within Chile's party system. On the left side of the spectrum, an increase in ideological diversity is apparent, with multiple parties emerging to represent different shades of left-leaning ideology. This fragmentation on the left indicates a trend of ideological diversification, where smaller and more ideologically specific groups continue to appear and position themselves relative to one another, resulting in a spectrum of left-leaning perspectives competing for influence.

In contrast, the right side of the spectrum exhibits a different pattern, where consolidation around fewer, more dominant entities is observed. This lack of fragmentation suggests a trend toward polarization on the right, as right-leaning parties seem less inclined to splinter into smaller factions, choosing instead to coalesce around core ideological positions. This creates a distinct asymmetry in the Chilean party system: while parties to the left of center demonstrate increasing ideological diversity, diluting their individual vote strength, those to the right appear to concentrate their support within a limited number of established parties, intensifying an internal ideological divide.

These findings suggest an evolving asymmetry within Chile's political system, where left-of-center politics are increasingly characterized by ideological fluidity and diversity, while right-of-center groups exhibit a more unified and polarized structure. This aligns with theoretical expectations of multiparty systems in which ideological

fragmentation often emerges in response to broader socio-political demands on the left, while the right may consolidate to counterbalance this ideological spread.⁴¹ Together, these trends reveal the complexity of ideological competition within Chile, illustrating that the Chilean political system may be undergoing a long-term shift toward asymmetric fragmentation, with implications for coalition dynamics and electoral strategies moving forward.

6. Final remarks

The results presented here demonstrate a comprehensive method to score and map ideological positions over time within complex party systems. This approach is grounded in foundational spatial theory, taking into account the dynamics of ideological positioning, strategic ambiguity, and multidimensional competition – factors essential to accurately capturing political landscapes over time. By anchoring our dataset in district-level data and enhancing it with synthetic estimations, this study represents the first effort to directly map the entirety of Chile's party system over time, rather than relying on insights derived from limited or isolated score subsets. Although the original data may have limitations, it is rooted in a thorough examination of public opinion polls, qualitative studies, and manifestos, ensuring consistency with established findings in the literature.

The combination of original and synthetic estimators provides a flexible and adaptive foundation for further refinement. Future iterations of this estimator could incorporate expert scores or additional external variables to reduce margins of error and avoid regions of lower density in the synthetic draws. This adaptability allows for contributions from the broader academic community, enhancing the reliability and applicability of this mapping approach, particularly for longitudinal analyses across varied electoral contexts.

The case study of Chile demonstrates the practical utility of this dataset and methodology, yielding nuanced insights into ideological trends within a multiparty system. Our findings address several key questions concerning ideological and organizational shifts in Chilean politics. Firstly, we identify signs of ideological fragmentation emerging as early as 2009 ([Figure 1](#)), especially within lower-level elections. This earlier fragmentation challenges the prevailing consensus that significant party system expansion began only after the 2015 electoral reform. The district-level granularity of our data allows for this broader perspective, capturing trends that may be obscured in higher-level analyses focused solely on legislative elections.

Secondly, we observe that left-leaning parties have shifted further to the left over time, indicating an ideological drift in this segment. Our regression analysis, highlighted in [Figure 2](#), reveals a statistically significant leftward shift among left-of-center parties, pointing to intensified ideological competition and diversification. This directional drift underscores a trend of ideological diffusion within the left, suggesting that left-of-center politics in Chile are characterized by increasing complexity and internal diversity.

Finally, we observe an asymmetrical pattern of ideological evolution, as shown in [Figure 3](#). While left-of-center parties increasingly fragment and diversify

ideologically, the right remains more consolidated and polarized. This discrepancy suggests a polarization trend on the right, where ideological cohesion appears stable, with a tendency toward a two-party structure. In contrast, the left exhibits greater ideological fluidity, expanding into a spectrum of nuanced positions. This asymmetry indicates that Chile's political landscape may be experiencing divergent ideological dynamics, with the left marked by fragmentation and the right by polarization.

These findings underscore the importance of a comprehensive, granular approach to ideological mapping, which allows researchers to capture subtle yet critical shifts that may be missed in higher-level analyses. The presented method not only addresses foundational theoretical questions but also offers a valuable tool for studying the effects of electoral reforms, ideological bias across districts, and the association between fragmentation and ideological positioning over time.

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24. Both pollsters have fielded polls since 1989, the year of the first election after the transition to democracy. Their full archives can be accessed here: <https://morichile.cl/> and here <https://cepchile.cl/>.
25. When discrepancies arose between survey and press data, survey data was prioritized given its statistical foundation. In cases of ambiguity or limited information, we applied an

averaging or triangulation method to reconcile differences, ensuring a balanced approach to scoring.

26. The Fixed Standard Deviation Model (FSDM) is designed to measure ideological positioning within coalitions by maintaining a consistent level of ideological dispersion. In this model, each party's score within a coalition is allowed to vary within a fixed, pre-defined standard deviation of 2 points from the coalition's mean ideological score. This choice is justified by historical data from CEP (Centro de Estudios Públicos) polls, which show that ideological scores for coalition-aligned parties in Chile typically remain within a 2-point range. These polls, conducted over multiple election cycles, consistently reveal that coalition parties exhibit a high degree of ideological discipline and cohesion, with limited deviations. By adopting a conservative fixed deviation, the FSDM captures realistic ideological shifts without overstating variability within coalitions, ensuring that fluctuations in party scores reflect genuine ideological shifts rather than random noise.

The Variable Standard Deviation Model (VSDM), by contrast, offers a more flexible approach by adjusting the standard deviation based on coalition composition and list size. In the VSDM, coalitions with a larger number of distinct parties are allowed greater variability in ideological positioning, reflecting the assumption that larger coalitions may accommodate a broader range of ideological viewpoints. For example, a coalition list containing four distinct parties would have a higher allowed standard deviation than a list with only two parties, capturing the potential for ideological diversity within larger coalitions. This model thus adapts to the idea that as coalitions increase in size and incorporate more parties, their internal ideological cohesion may become more diffuse. The VSDM approach is particularly useful for exploring how coalition dynamics might fluctuate over time, as the ideological spread within a coalition could vary with changes in the number of parties or shifts in coalition discipline.

Although the code and data associated with this publication include results for both models, we have chosen to present the FSDM results in the main text for two reasons. First, the fixed 2-point deviation aligns with historical data on coalition cohesion in Chile, as evidenced by CEP polls that suggest ideological scores within coalition parties rarely deviate by more than 2 points. This stability justifies the use of a conservative fixed standard deviation, which avoids inflating perceived ideological differences within coalitions. Second, the FSDM helps prevent overfitting, especially in cases where data availability for specific coalition parties is limited. A variable standard deviation, while adaptable, could distort results by magnifying minor variations in smaller or newer coalition parties that lack comprehensive data. By maintaining a fixed 2-point deviation, the FSDM provides a robust, generalizable framework for longitudinal analysis.

To validate this choice, we conducted diagnostic checks for both models, as shown in Figures A3 to A8 for the FSDM and Figures B3 to B8 for the VSDM. Additionally, Figure A21 compares the density distributions of both models, illustrating that the FSDM produces a more conservative, cohesive estimate of coalition ideology, whereas the VSDM introduces greater dispersion based on coalition size. This comparative analysis confirms that the FSDM is a reliable, stable model for capturing ideological shifts within Chile's coalition-based political system, especially for applications requiring consistent longitudinal tracking.

27. Their full archives can be accessed here: <https://latercera.cl/> and here <https://elmercurio.cl/>.
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29. The introduction of random noise to the original scores simulates minor fluctuations that reflect natural ideological variation within each category while preserving core ideological structure. This approach follows established practices in synthetic data

generation, where noise is added to capture realistic but minimal deviations. Here, we opted for a conservative noise range between -0.25 and 0.25 after testing multiple ranges. Our tests (see Figures A23 to A26 in the Appendix) confirmed that deviations within this range do not significantly impact ideological placements, with noticeable shifts only observed when noise exceeded ± 1.0 . Notably, a noise level of 1.0 represents a 10% shift on our 0–10 scale, which users can toggle within the provided code script to adjust for different levels of variation. These findings, evident in the appended figures, show that ideological positioning remains stable under low-noise scenarios, only diverging meaningfully with noise around or above 1.0 . This conservative noise calibration ensures replicability and maintains dataset integrity, supporting visual pattern recognition rather than exact positional adjustments. This balanced approach addresses reviewer concerns, adding realism while safeguarding the ideological structure of the data.

30. No significant difference was detected between simulations of $N = 100$ and simulations of $N = 1,000$, $N = 10,000$, and $N = 100,000$. The script is standardized at 1,000 simulations, but this can be toggled to test different levels using the code script.
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32. Figure B1 and B2 in the Appendice show the same for the VSDM.
33. Figure B3 through B8 in the Appendice show the same for the VSDM.
34. Figure B9 and B10 in the Appendice show the same for the VSDM.
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39. An example of the ideological trajectory of some of the main political parties is presented in Figure A11 to Figure A12 in the Appendice.
40. A jitter factor is incorporated to represent the variation across different competitions held within the same year, particularly for independents, to emphasize their relatively minor weight in the overall electoral landscape. Users can toggle the jitter factor using the code script, yet it will not result in significant underlying change, as it is just intended enhance visualization of ideological competition differences. Note that the colors correspond to the ideological categories established in Table 1.
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